Chapter 4

Probability

In Chapter 3 we explained how to use sample statistics as point estimates of population parameters.

In this chapter and in Chapters 5 and 6 we present the fundamental concepts about probability that are needed to understand how we make such statistical inferences. We begin our discussions in this chapter by considering rules for calculating probabilities. In order to illustrate some of the concepts in this chapter, we will introduce a new case.

Chapter Outline

4.1 The Concept of Probability - Probability and Sample Spaces

4.2 Sample Spaces and Events

4.3 Some Elementary Probability Rules

4.4 Conditional Probability and Independence

4.1 The Concept of Probability - Probability and Sample Spaces

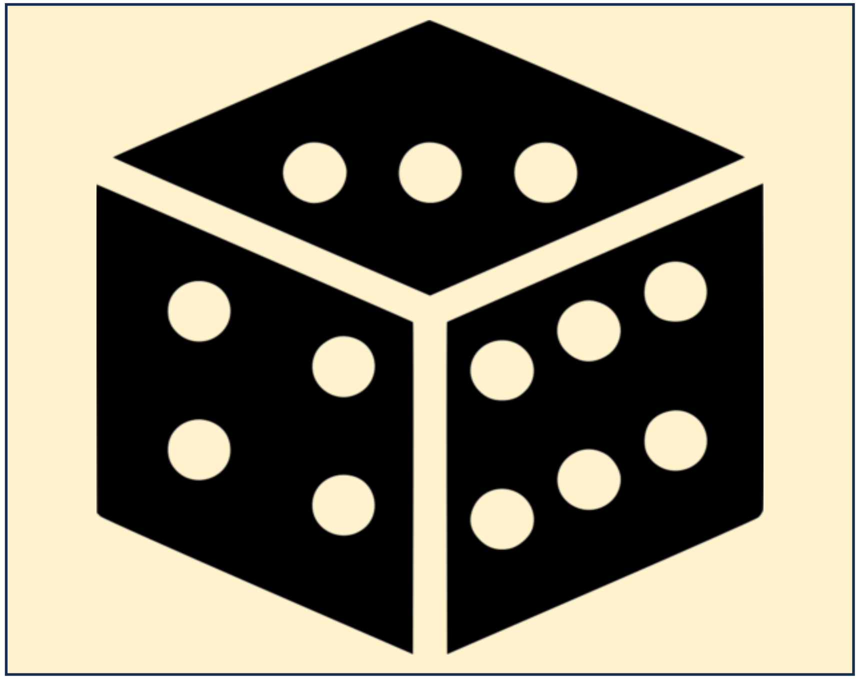
- An **experiment** is any process of observation with an uncertain outcome.

- The **sample space** of an experiment is the set of all possible outcomes for the experiment.

- The possible outcomes are sometimes called the **experimental outcomes or sample space outcomes.**

- **Probability** is a measure of the chance (possibility) that an experimental outcome will occur when an experiment is carried out.

Example – throw a dice



- It is an experiment, with an uncertain outcome.
- The possible outcomes (experimental outcomes / sample space outcomes) are?
- The sample space is?
- Probability of outcome “one spot”, “two spots”,…, “six spots” is?

Probability

- If E is a sample space outcome, then $P(E)$ denotes the probability that E will occur and:
- **Conditions:**
  - $0 \leq P(E) \leq 1$ such that:
    - If E can never occur, then $P(E) = 0$
    - If E is certain to occur, then $P(E) = 1$
  - The probabilities of all the sample space outcomes must sum to 1.

Example: Let A, B, C, D, and E be sample space outcomes forming a sample space. Suppose that

$$P(A) = 0.2, P(B) = 0.15, P(C) = 0.3, P(D) = 0.2.$$

What is $P(E)$? Explain how you got your answer.

Assigning Probabilities to Sample Space Outcomes

(1) Classical method

- Can be used when the sample space outcomes are equally likely.

(2) Relative frequency method

- Using the long run relative frequency.

(3) Subjective method

- Assessment based on experience, expertise or intuition.

(1) Classical method

- **Example1:** consider the experiment of **tossing a fair coin**. Here, there are two equally likely sample space outcomes — head (H) and tail (T). Therefore, logic suggests that the probability of observing a head, denoted $P(H)$, is ½ = 0.5, and that the probability of observing a tail, denoted $P(T)$, is also ½ = 0.5.

- **Example 2:** consider the experiment of **rolling a fair die**. It would seem reasonable to think that the six sample space outcomes 1, 2, 3, 4, 5, and 6 are equally likely, and thus each outcome is assigned a probability of 1/6. If $P(1)$ denotes the probability that one dot appears on the upward face of the die, then $P(1) = 1/6$. Similarly, $P(2) = 1/6$, $P(3) = 1/6$, $P(4) = 1/6$, $P(5) = 1/6$, and $P(6) =1/6$.

(2) Relative frequency method

Background: Consider tossing a fair coin — theoretically, a coin such that the probability of its upward face showing as a head is 0.5. If we get 6 heads in the first 10 tosses, then the relative frequency, or fraction, of heads is 6 / 10 = 0.6. If we get 47 heads in the first 100 tosses, the relative frequency of heads is 47 / 100 = 0.47. If we get 5,067 heads in the first 10,000 tosses, the relative frequency of heads is 5,067 / 10,000 = 0.5067. Note that the relative frequency of heads is approaching (that is, getting closer to) 0.5. The long run relative frequency interpretation of probability says that, if we tossed the coin an indefinitely large number of times (that is, a number of times approaching infinity), the relative frequency of heads obtained would approach 0.5. Of course, in actuality, it is impossible to toss a coin (or perform any experiment) an indefinitely large number of times. Therefore, a relative frequency interpretation of probability is an idealization.

- **For example,** to estimate the probability that a randomly selected consumer prefers Coca-Cola to all other soft drinks, we perform an experiment in which we ask a randomly selected consumer for his or her preference. There are two possible experimental outcomes: “prefers Coca-Cola” and “does not prefer Coca-Cola.”
- We might perform the experiment, say, 1,000 times by surveying 1,000 randomly selected consumers.
- Then, if 140 of those surveyed said that they prefer Coca-Cola, we would estimate the probability that a randomly selected consumer prefers Coca-Cola to all other soft drinks to be 140 / 1,000 = 0.14.
- This is an example of the relative frequency method of assigning probability.

(3) Subjective method

- When we use experience, intuitive judgement, or expertise to assess a probability, we call this **the subjective method** of assigning probability.

- For instance, when the company president estimates that the probability of a successful business venture is 0.7.

- This may mean that, if business conditions similar to those that are about to be encountered could be repeated many times, then the business venture would be successful in 70 percent of the repetitions.

4.2 Sample Spaces and Events

- **Sample Space:** The set of all possible experimental outcomes.
- **Sample Space Outcomes:** The experimental outcomes in the sample space.
- **Event:** A set of sample space outcomes (a **subset** of sample space).
- **Probability:** The probability of an event is the sum of the probabilities of the sample space outcomes that correspond to the event.

For example, if we consider the experiment of tossing a fair die, the event "*at least five spots will show on the upward face of the die*" consists of the sample space outcomes 5 and 6. That is, the event "at least five spots will show on the upward face of the die" will occur if and only if one of the sample space outcomes 5 or 6 occurs.

Roll dice

In this experiment

- **Sample Space**: 1, and 2, and 3, and 4, and 5, and 6.
- **Sample Space Outcomes**: 1, or 2, or 3, or 4, or 5, or 6.
- **Event**: "at least five spots will show on the upward face of the die" is an event.
- **Probability**: The probability of the above event is 2 / 6 = 0.33.

Example 4.1 Boys and girls

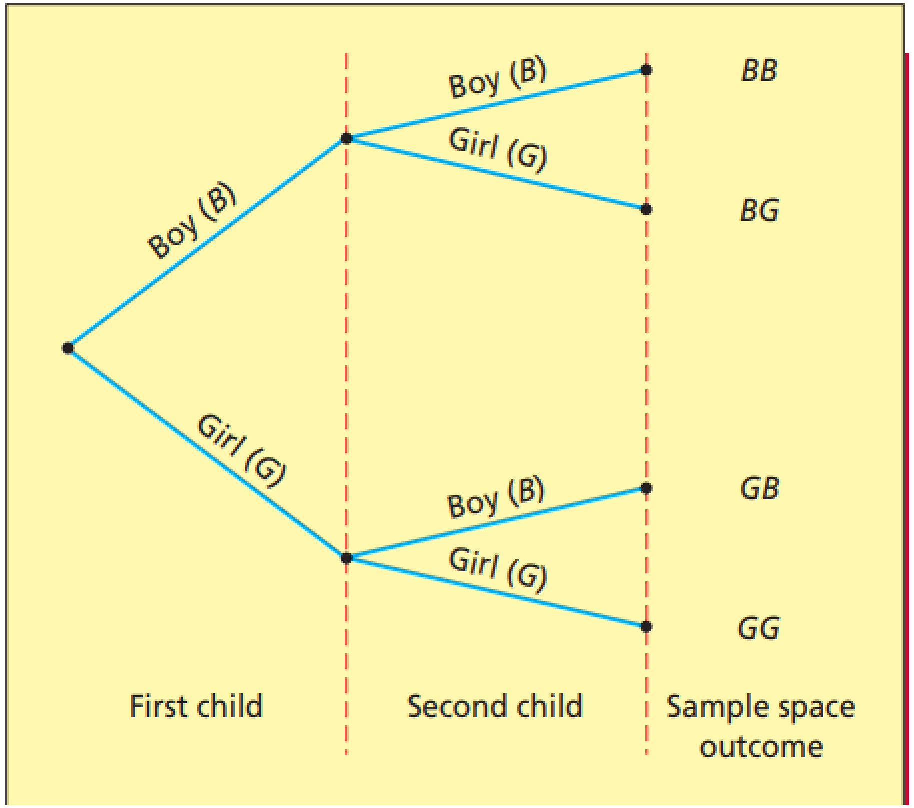
A couple plans to have two children. We consider the experiment of having two children. We let B denote that a child is a boy and G denote that a child is a girl. Then, it is useful to construct the tree diagram shown in Figure 4.1. Each branch of the tree leads to a sample space outcome. These outcomes are listed at the right ends of the branches. We see that there are four sample space outcomes. Therefore, the sample space (that is, the set of all the sample space outcomes) is:

$$BB \quad BG \quad GB \quad GG$$

Intuitively, each of the sample space outcomes is equally likely. That is, this implies that

$$P(BB) = P(BG) = P(GB) = P(GG) = \frac{1}{4}$$

FIGURE 4.1 A Tree Diagram of the Genders of Two Children



Therefore:

1. The probability that the couple will have two boys is

$$P(BB) = \frac{1}{4}$$

2. The probability that the couple will have one boy and one girl is

$$P(BG) + P(GB) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

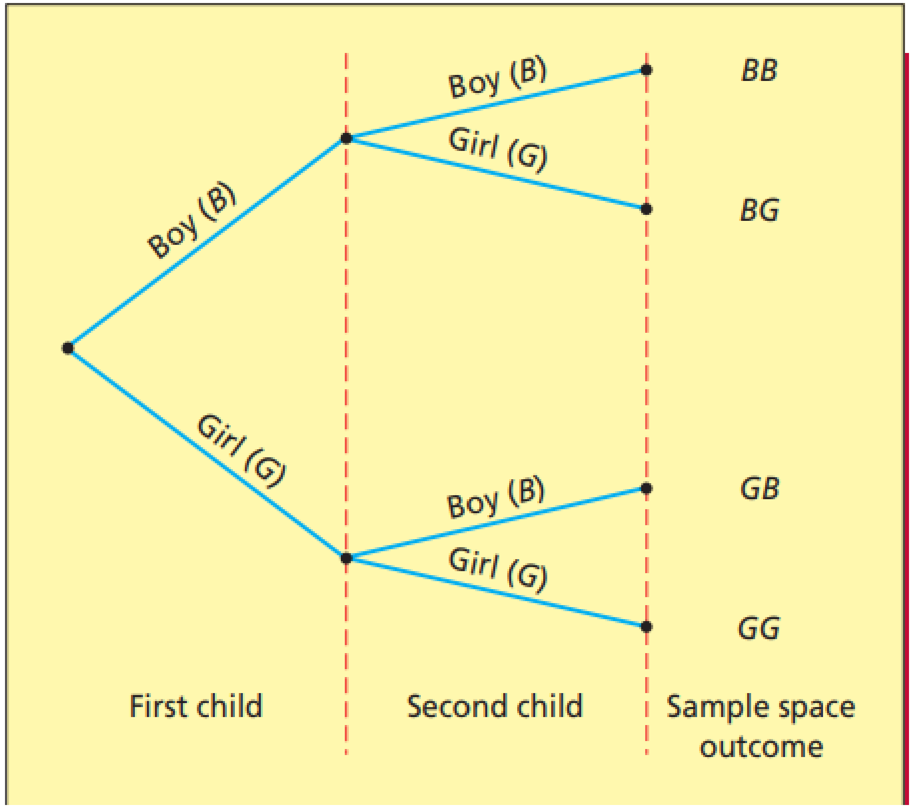
3. The probability that the couple will have two girls is

$$P(GG) = \frac{1}{4}$$

4. The probability that the couple will have at least one girl is

$$P(BG) + P(GB) + P(GG) = \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = \frac{3}{4}$$

FIGURE 4.1 A Tree Diagram of the Genders of Two Children

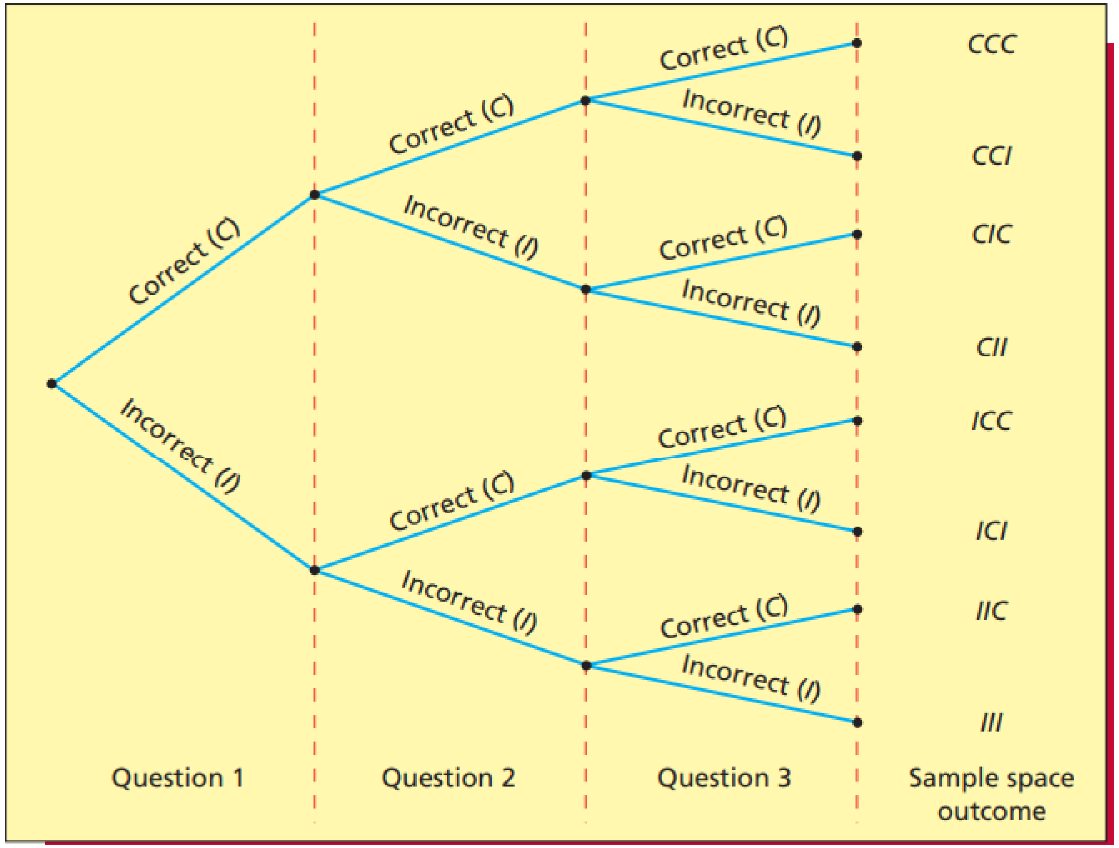


Example 4.2 Pop quizzes

A pop quiz that consists of three true–false questions. If we consider our experiment to be answering the three questions, each question can be answered correctly (C) or incorrectly (I). Figure 4.2 depicts a tree diagram of the sample space outcomes for the experiment. The tree diagram has eight different branches, and the eight sample space outcomes are listed at the ends of the branches. We see that the sample space is

$$CCC \quad CCI \quad CIC \quad CII \quad ICC \quad ICI \quad IIC \quad III$$

FIGURE 4.2 A Tree Diagram of Answering Three True–False Questions



Suppose that the student was totally unprepared for the quiz and had to blindly guess the answer to each question. Intuitively, this would say that each of the eight sample space outcomes is equally likely to occur. That is

$$P(CCC) = P(CCI) = \cdots = \frac{1}{8}$$

Therefore:

1. The probability that the student will get all three questions correct is

$$P(CCC) = \frac{1}{8}$$

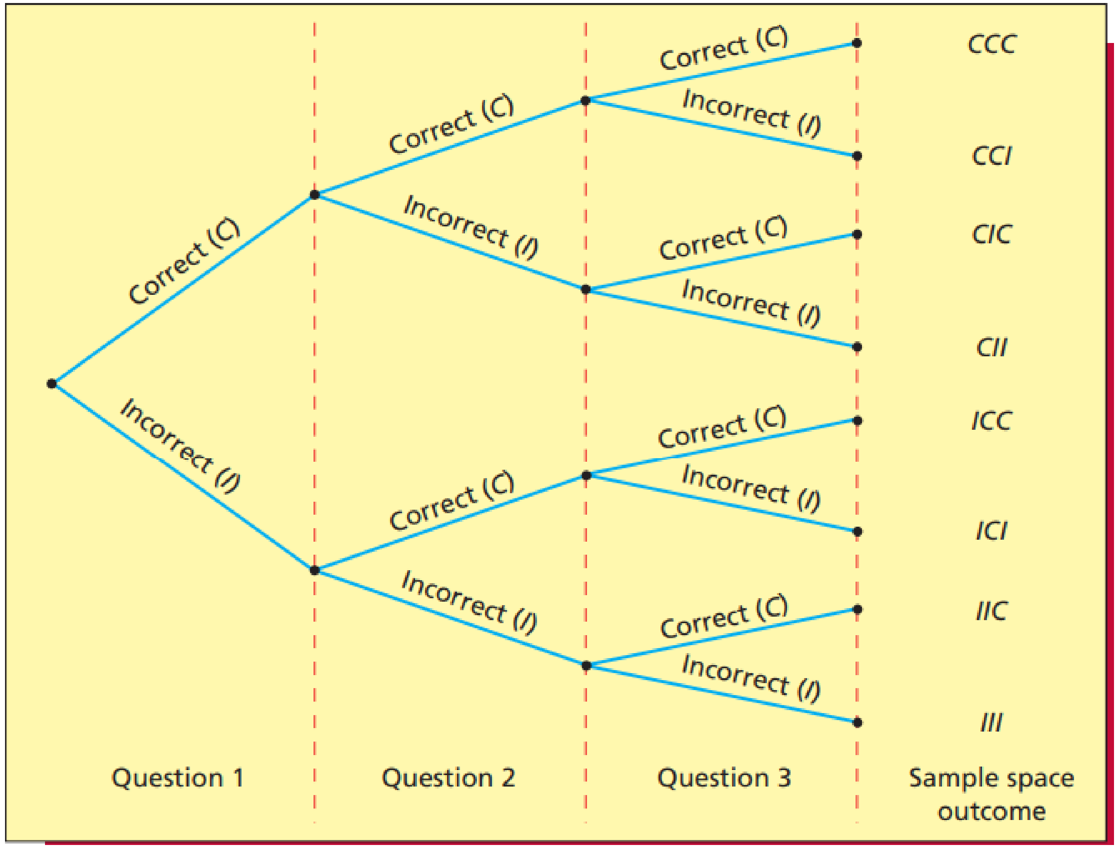
2. The probability that the student will get exactly two questions correct is

$$P(CCI) + P(CIC) + P(ICC) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

3. The probability that the student will get all three questions incorrect is

$$P(III) = \frac{1}{8}$$

FIGURE 4.2 A Tree Diagram of Answering Three True–False Questions



4. The probability that the student will get exactly one question correct is

$$P(CII) + P(ICI) + P(IIC) = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

5. The probability that the student will get at least two questions correct is

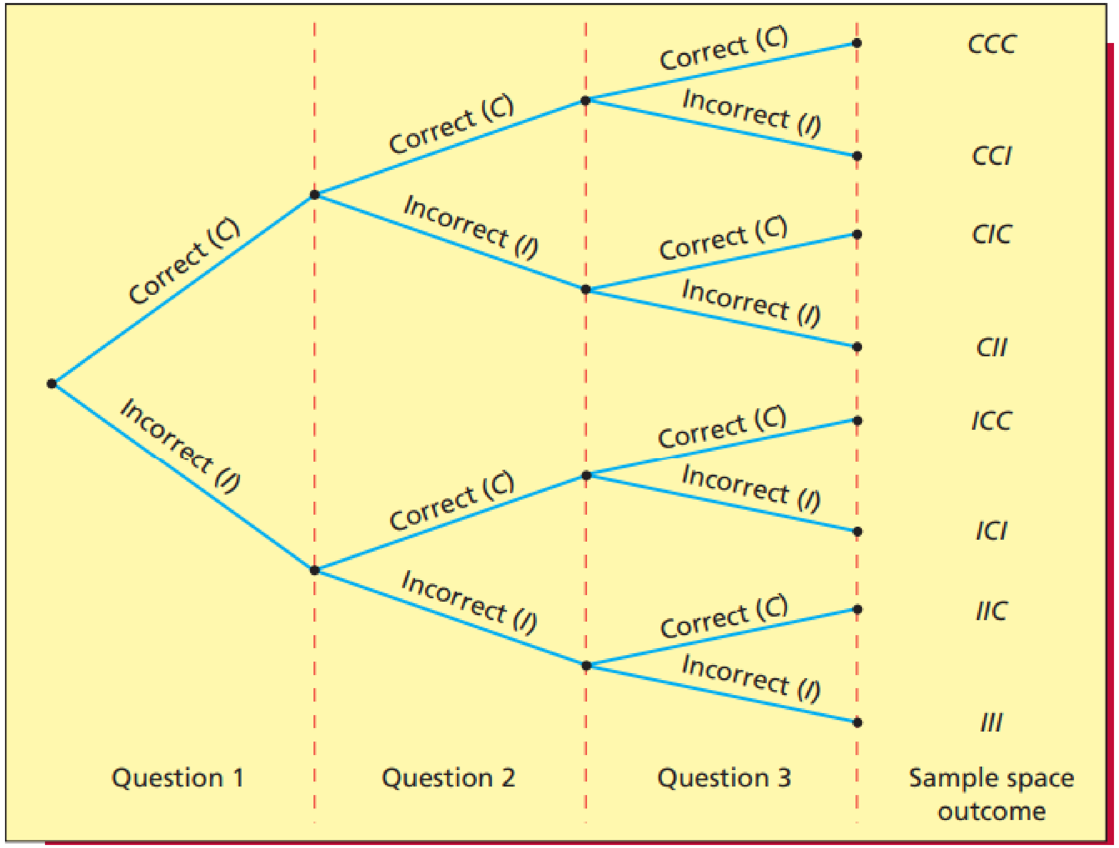
$$P(CCC) + (CCI) + P(CIC) + P(ICC)$$

$$= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$$

6. The probability that the student will get exactly one question correct equals the ratio

$$\frac{\text{the number of sample space outcomes resulting in one correct answer}}{\text{the total number of sample space outcomes}} = \frac{3}{8}$$

FIGURE 4.2 A Tree Diagram of Answering Three True–False Questions



Finding Simple Probabilities

In general, when a sample space is finite we can use the following method for computing the probability of an event.

If all of the sample space outcomes are ***equally likely***, then the probability that an event will occur is equal to the ratio

$$\frac{\text{the number of sample space outcomes that correspond to the event}}{\text{the total number of sample space outcomes}}$$

End of the first two classes

Example 4.3 Choosing a CEO

A company is choosing a new chief executive officer (CEO) from the list of 4 candidates—Adams, Chung, Hill, and Rankin. If we consider our experiment to be making a final choice of the company's CEO, then the experiment's sample space consists of the four possible outcomes:

- A - Adams will be chosen as CEO.
- C - Chung will be chosen as CEO.
- H - Hill will be chosen as CEO.
- R - Rankin will be chosen as CEO.

Next, suppose that industry analysts feel (subjectively) that the probabilities that Adams, Chung, Hill, and Rankin will be chosen as CEO are 0.1, 0.2, 0.5, and 0.2, respectively. That is, in probability notation

$$P(A) = 0.1, P(C) = 0.2, P(H) = 0.5, P(R) = 0.2$$

Also, suppose only Adams and Hill are internal candidates (they already work for the company). Letting *Int* denote the event that "an internal candidate will be selected for the CEO position," then *Int* consists of the sample space outcomes A and H (that is, *Int* will occur **if and only if** either of the sample space outcomes A or H occurs). It follows that $P(Int) = P(A) + P(H) = 0.1 + 0.5 = 0.6$. This says that the probability that an internal candidate will be chosen to be CEO is 0.6.

Note: If we had ignored the fact that sample space outcomes are not equally likely, we might have tried to calculate $P(Int)$ as follows:

$$\cancel{P(Int) = \frac{\text{the number of internal candidates}}{\text{the total number of candidates}} = \frac{2}{4} = 0.5}$$

This result would be incorrect. Because the sample space outcomes are not equally likely, we have seen that the correct value of $P(Int)$ is 0.6, not 0.5.

Example 4.4 The Crystal Cable Case: Market Penetration

- A cable company can compare the number of households it actually serves to the number of households its current transmission lines could reach (without extending the lines).
- The number of households that the cable company’s lines could reach is called its number of **cable passings**, while the ratio of the number of households the cable company actually serves to its number of cable passings is called the company’s cable penetration.
- There are various types of **cable passings**— one for cable television, one for cable Internet, one for cable phone, and others.

- In a recent quarterly report, Crystal Cable reported that it had 12.4 million cable television customers and 27.4 million cable passings. Consider randomly selecting one of Crystal's cable passings. That is, consider selecting one cable passing by giving each and every cable passing the same chance of being selected. Let ***A*** be the event that the randomly selected cable passing has Crystal's cable television service. Then, because the sample space of this experiment consists of 27.4 million equally likely sample space outcomes (cable passings), it follows that

$$P(A) = \frac{\text{the number of cable passings that have Crystal's cable television service}}{\text{the total number of cable passings}} = \frac{12.4}{27.4} = 0.455$$

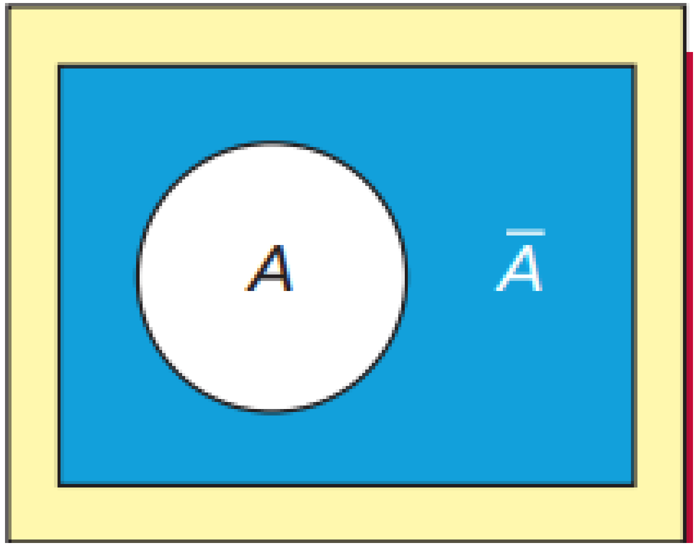
This probability is Crystal's cable television penetration and says that the probability that a randomly selected cable passing has Crystal's cable television service is 0.45. That is, 45 percent of Crystal's cable passings have Crystal's cable television service.

4.3 Some elementary probability rules

- Complement
- Union
- Intersection
- Addition
- Conditional probability

Complement

- **The complement** of an event A is denoted as $\bar{A}$. It is the set of all sample space outcomes NOT in A



- $P(\bar{A}) + P(A) = 1$

The Rule of Complements

Consider an event A. Then, the probability that A will not occur is

$$P(\bar{A}) = 1 - P(A)$$

Example 4.5 The Crystal Cable Case: Market Penetration

Recall from Example 4.4 that the probability that a randomly selected cable passing has Crystal's cable television service is 0.45. It follows that the probability of the complement of this event (that is, the probability that a randomly selected cable passing does not have Crystal's cable television service) is $1 - 0.45 = 0.55$.

Intersection of two events

Given two events A and B, the **intersection of *A* and *B*** is the event that occurs if both A **and** B simultaneously occur. The intersection is denoted by $A \cap B$. Furthermore, $P(A \cap B)$ denotes **the probability that both *A* and *B* will simultaneously occur.**

$P(A \cap B)$ is the **joint probability** of A and B both occurring together

Example 4.6 The Crystal Cable Case: Market Penetration

Recall from Example 4.4 that Crystal Cable has 27.4 million cable passings. Consider randomly selecting one of these cable passings, and define the following events:

$A \equiv$ the randomly selected cable passing has Crystal's cable television service.

$\overline{A} \equiv$ the randomly selected cable passing does not have Crystal's cable television service.

$B \equiv$ the randomly selected cable passing has Crystal's cable Internet service.

$\overline{B} \equiv$ the randomly selected cable passing does not have Crystal's cable Internet service.

$A \cap B \equiv$ the randomly selected cable passing has both Crystal's cable television service and Crystal's cable Internet service.

$A \cap \overline{B} \equiv$ the randomly selected cable passing has Crystal's cable television service and does not have Crystal's cable Internet service.

$\overline{A} \cap B \equiv$ the randomly selected cable passing does not have Crystal's cable television service and does have Crystal's cable Internet service.

$\overline{A} \cap \overline{B} \equiv$ the randomly selected cable passing does not have Crystal's cable television service and does not have Crystal's cable Internet service.

TABLE 4.1 A Contingency Table Summarizing Crystal's Cable Television and Internet Penetration (Figures In Millions Of Cable Passings)

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

Using this table, we can calculate the following probabilities, each of which describes some aspect of Crystal's cable penetrations:

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

1 Because 12.4 million out of 27.4 million cable passings have Crystal's cable television service, A, then

$$P(A) = \frac{12.4}{27.4} = .45$$

This says that 45 percent of Crystal's cable passings have Crystal's cable television service (as previously seen in Example 4.4).

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

2 Because 9.8 million out of 27.4 million cable passings have Crystal's cable Internet service, B, then

$$P(B) = \frac{9.8}{27.4} = .36$$

This says that 36 percent of Crystal's cable passings have Crystal's cable Internet service.

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

3 Because 6.5 million out of 27.4 million cable passings have Crystal's cable television service and Crystal's cable Internet service, $A \cap B$, then

$$P(A \cap B) = \frac{6.5}{27.4} = .24$$

This says that 24 percent of Crystal's cable passings have both of Crystal's cable services.

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

4 Because 5.9 million out of 27.4 million cable passings have Crystal's cable television service, but do not have Crystal's cable Internet service, $A \cap \overline{B}$, then

$$P(A \cap \overline{B}) = \frac{5.9}{27.4} = .22$$

This says that 22 percent of Crystal's cable passings have only Crystal's cable television service.

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

5 Because 3.3 million out of 27.4 million cable passings do not have Crystal's cable television service, but do have Crystal's cable Internet service, $\overline{A} \cap B$, then

$$P(\overline{A} \cap B) = \frac{3.3}{27.4} = .12$$

This says that 12 percent of Crystal's cable passings have only Crystal's cable Internet service.

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

6 Because 11.7 million out of 27.4 million cable passings do not have Crystal's cable television service and do not have Crystal's cable Internet service, $\overline{A} \cap \overline{B}$, then

$$P(\overline{A} \cap \overline{B}) = \frac{11.7}{27.4} = .43$$

This says that 43 percent of Crystal's cable passings have neither of Crystal's cable services.

Union of two events

Given two events A and B, the **union of *A* and *B*** is the event that occurs if A **or** B **or** both occur. The union is denoted by $A \cup B$. Furthermore, $P(A \cup B)$ denotes **the probability that *A* or *B* or both will occur.**

Have A, or have B, or have A and B

Intersection vs. Union

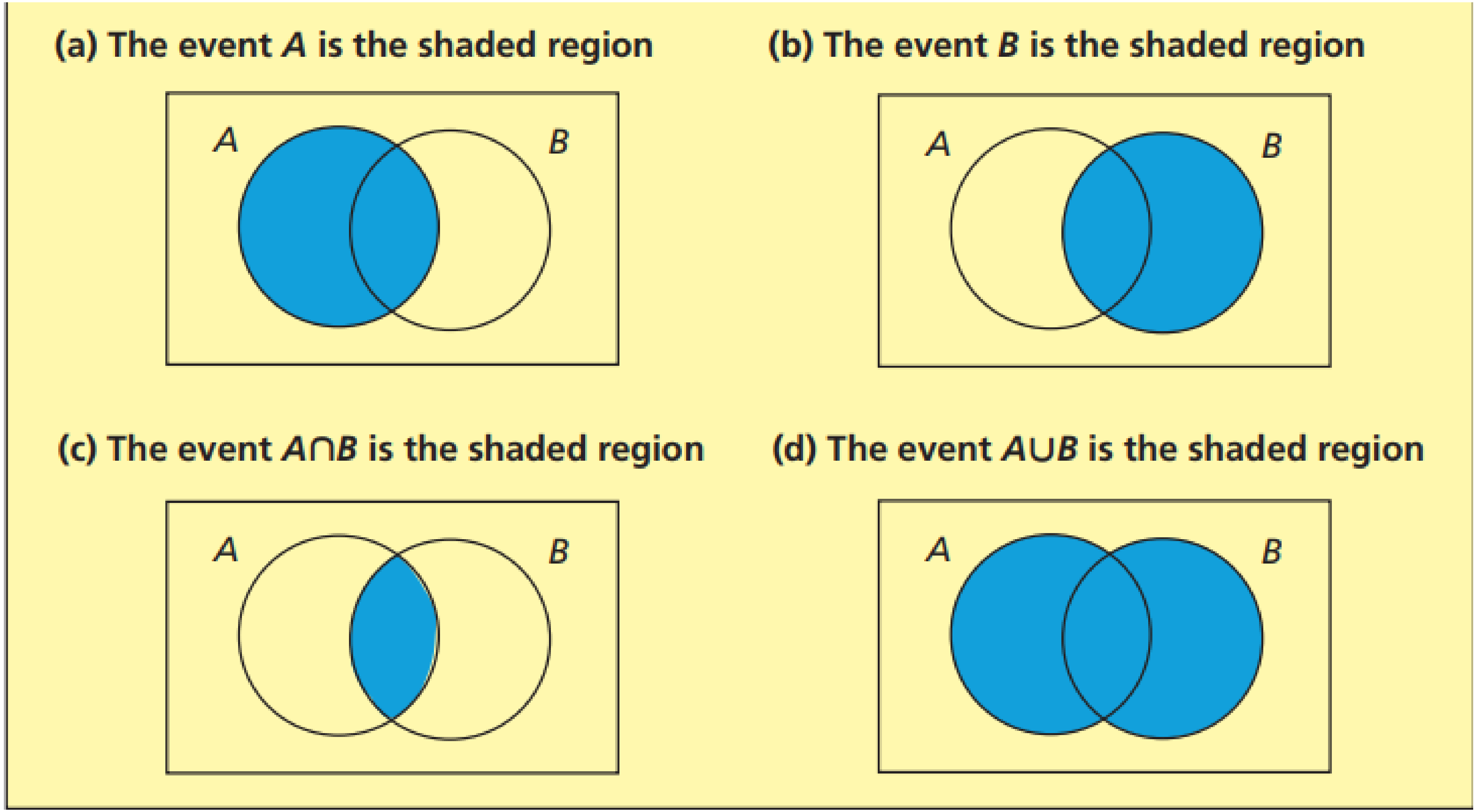
Given two events A and B, the **intersection of A and B** is the event that occurs if both A **and** B simultaneously occur. The intersection is denoted by $A \cap B$. Furthermore, $P(A \cap B)$ denotes **the probability that both A and B will simultaneously occur.**

Given two events A and B, the **union of A and B** is the event that occurs if A **or** B **or** both occur. The union is denoted by $A \cup B$. Furthermore, $P(A \cup B)$ denotes **the probability that A or B or both will occur.**

- The **intersection** of A and B are elementary events that belong to both A **and** B. Written as $A \cap B$.

- The **union** of A and B are elementary events that belong to either A **or** B **or** both. Written as $A \cup B$.

FIGURE 4.4 Venn Diagrams Depicting the Events *A*, *B*, $A \cap B$, and $A \cup B$



Example 4.7 The Crystal Cable Case: Market Penetration

Consider randomly selecting one of Crystal's 27.4 million cable passings, and define the event

$A \cup B \equiv$ the randomly selected cable passing has Crystal's cable television service **or** Crystal's cable Internet service (**or** both) — that is, has at least one of the two services.

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\bar{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\bar{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

Looking at Table 4.1 (above), we see that

$$P(A \cup B) = \frac{5.9+3.3+6.5}{27.4} = \frac{15.7}{27.4} = 0.57$$

$$P(A) + P(B) = \frac{12.4}{27.4} + \frac{9.8}{27.4} = \frac{15.7}{27.4} = 0.81$$

$$P(A \cap B) = \frac{6.5}{27.4} = 0.24$$

It can be seen that

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

The Addition Rule

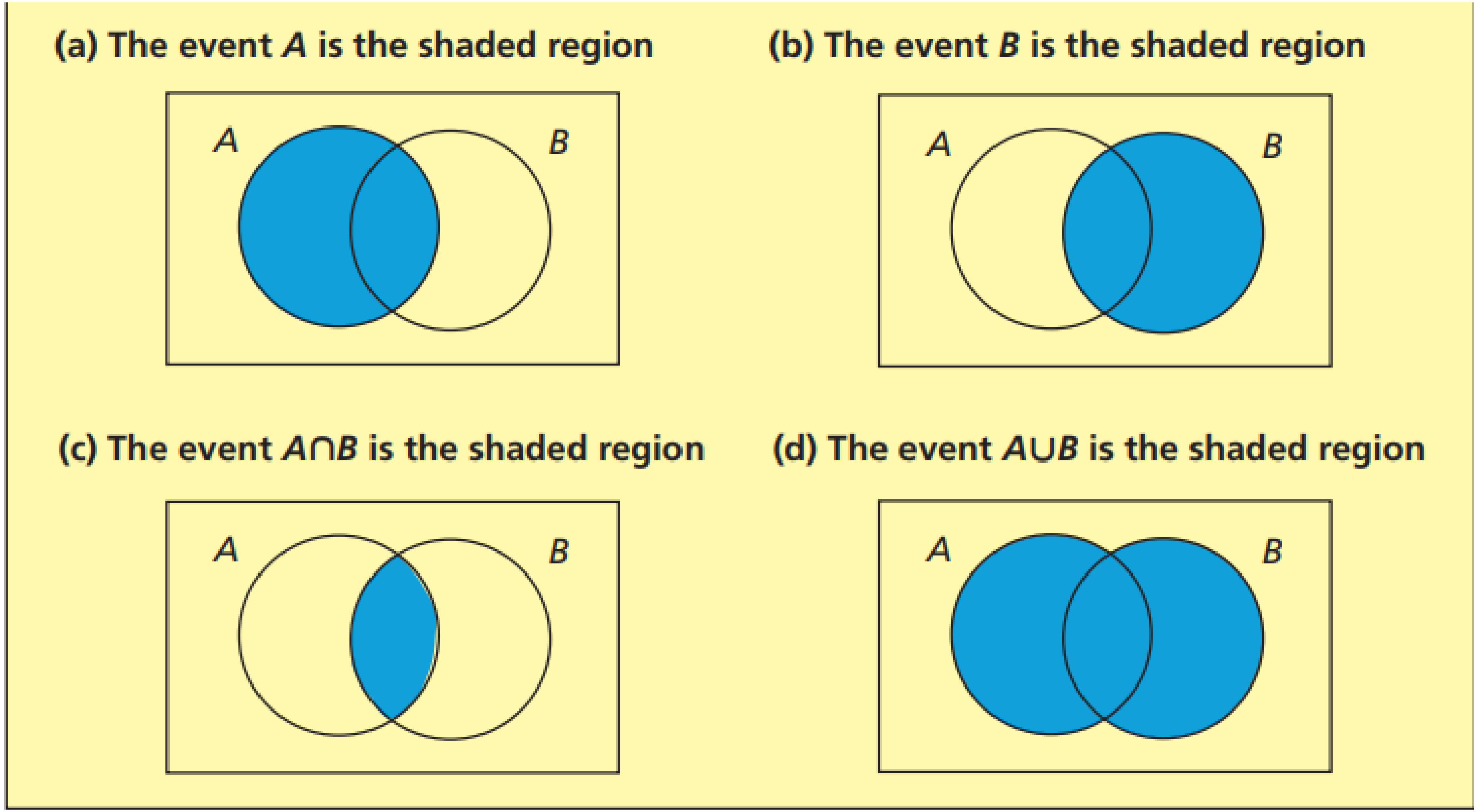
Let A and B be events. Then, the probability that A or B (or both) will occur is

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

where $P(A \cap B)$ is the **joint probability** of A and B both occurring together

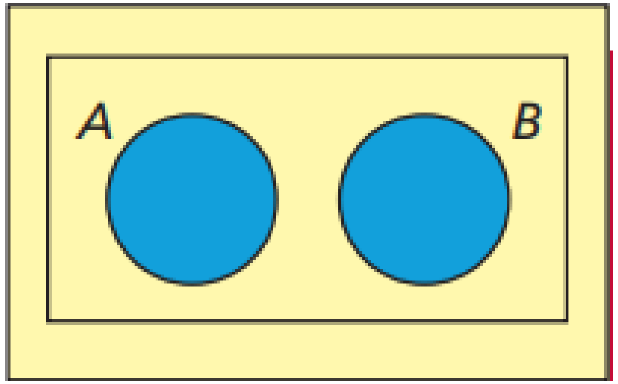
Figure 4.4 shows that when we compute $P(A) + P(B)$, we are counting each of the sample space outcomes in $A \cap B$ twice. We correct for this by subtracting $P(A \cap B)$.

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$



Mutually Exclusive Events

Two events A and B are **mutually exclusive** if they have no sample space outcomes in common. In this case, the events A and B **cannot** occur simultaneously, and thus $P(A \cap B) = 0$.



Addition rules

- If A and B are **mutually exclusive**, then the probability that A or B (the union of A and B) will occur is

$$P(A \cup B) = P(A) + P(B)$$

- If A and B are ***not mutually exclusive***:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

where $P(A \cap B)$ is the **joint probability** of A and B both occurring together.

Example 4.8 - 4.9 Selecting Playing Cards

Consider randomly selecting a card from a standard deck of 52 playing cards. We define the following events:

J ≡ the randomly selected card is a jack.

Q ≡ the randomly selected card is a queen.

R ≡ the randomly selected card is a red card (that is, a diamond or a heart).

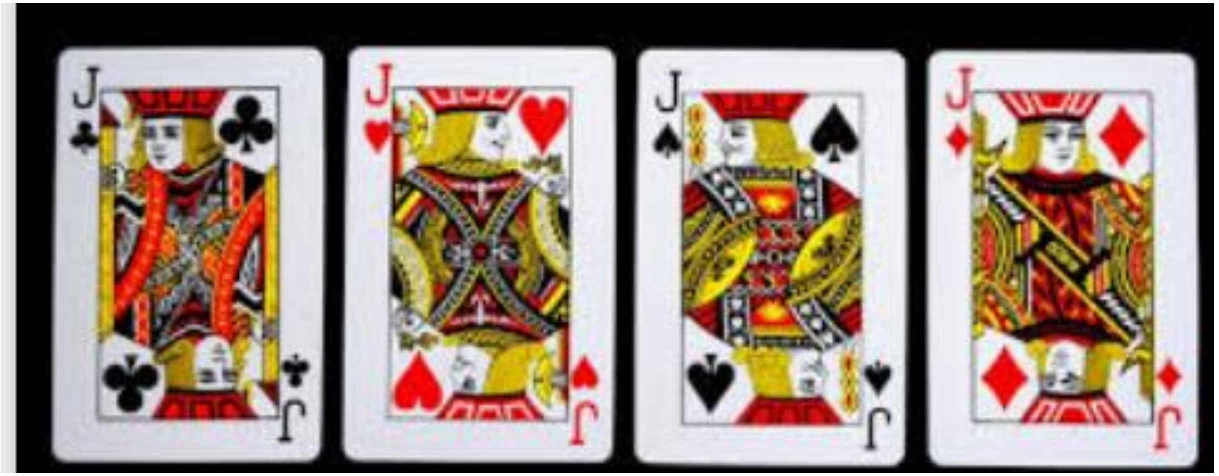
There is no card that is both a jack and a queen, the events J and Q are mutually exclusive. There are two cards that are both jacks and red cards—the jack of diamonds and the jack of hearts—the events J and R are not mutually exclusive.

Because there are four jacks, four queens, and 26 red cards, we have $P(J) = \frac{4}{52}, P(Q) = \frac{4}{52},$ and $P(R) = \frac{26}{52}$. It follows that the probability that the randomly selected card is a jack or a queen is

$$P(J \cup Q) = P(J) + P(Q) = \frac{4}{52} + \frac{4}{52} = \frac{8}{52} = \frac{2}{13}$$

and the probability that the randomly selected card is a jack or a red card or both is

$$P(J \cup R) = P(J) + P(R) - P(J \cap R) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13}$$

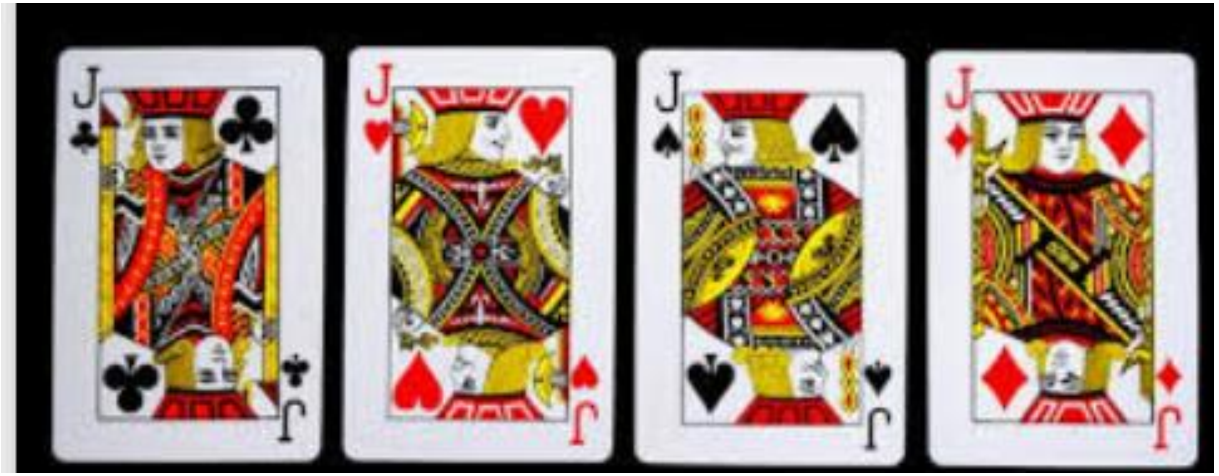


In fact, there are 4 jacks and 24 red cards that are not jacks (the 26 red cards minus the 2 red jacks). Therefore, because there are a total of 28 cards corresponding to the event $P(J \cup R)$, it follows that

$$P(J \cup R) = \frac{28}{52} = \frac{7}{13}$$

This confirms the equation:

$$P(J \cup R) = P(J) + P(R) - P(J \cap R)$$



There are 4 jacks, 4 queens, and 22 red cards that are not jacks or queens (the 26 red cards minus the 2 red jacks and the 2 red queens). Therefore, because there are a total of 30 cards corresponding to the event $P(J \cup Q \cup R)$, it follows that

$$P(J \cup Q \cup R) = \frac{30}{52} = \frac{15}{26}$$

For an arbitrary group of events—A_1, A_2, . . . , A_N, the probability that A_1 or A_2 or . . .or A_N occurs (that is, the probability that at least one of the events occurs) as $P(A_1 \cup A_2 \cup A_3 \ldots \cup A_N)$. Although there is a formula for this probability, it is quite complicated and we will not present it.

The Addition Rule for N Mutually Exclusive Events

The events $A_1, A_2, \ldots, A_N$ are mutually exclusive if no two of the events have any sample space outcomes in common. In this case, no two of the events can occur simultaneously, and

$$P(A_1 \cup A_2 \cup A_3 \ldots \cup A_N) = P(A_1) + P(A_2) \ldots + P(A_N)$$

As an example of using this formula, again consider the playing card situation and the events J and Q. If we define the event

$$K \equiv \text{the randomly selected card is a king}$$

then the events J, Q, and K are mutually exclusive. Therefore,

$$P(J \cup Q \cup K) = P(J) + P(Q) + P(K)$$
$$= \frac{4}{52} + \frac{4}{52} + \frac{4}{52} = \frac{12}{52} = \frac{3}{13}$$

4.4 Conditional Probability and Independence

- The probability of an event A, given that the event B has occurred, is **called the conditional probability of A given B**
  - Denoted as $P(A|B)$
- Further, $\boxed{P(A|B) = \frac{P(A \cap B)}{P(B)}}$
  - $P(B) \neq 0$
- Likewise, $\boxed{P(B|A) = \frac{P(A \cap B)}{P(A)}}$

- The equations are important.
- We will demonstrate them in the following example.

Interpretation

- Restrict sample space to just event B.
- The conditional probability $P(A|B)$ is the chance of event A occurring in this **new** sample space. Pronounced "the probability of A given B."
- In other words, if B occurred, then what is the chance of A occurring.

General Multiplication Rule

Given any two events, A and B

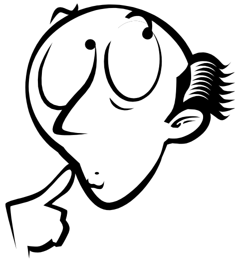
$$\boxed{P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)}$$

- The probability of event A occurs is $P(A)$.
- After A occurs, the probability of the event B occurs is $P(B|A)$.
- Therefore, the probability of both A and B occur is $P(A)$ time $P(B|A)$

Table 4.1 (below) summarise data concerning Crystal cable's 27.4 million cable passings. Suppose that we randomly select a cable passing and that the chosen cable passing reports that it has Crystal's cable Internet service. Given this new information, we wish to find the probability that the cable passing has Crystal's cable television service. This new probability is called a conditional probability.

Events	Has Cable Internet Service, B	Does Not Have Cable Internet Service, $\overline{B}$	Total
Has Cable Television Service, A	6.5	5.9	12.4
Does Not Have Cable Television Service, $\overline{A}$	3.3	11.7	15.0
Total	9.8	17.6	27.4

Understand the question.



- The randomly selected cable passing has Internet service (Given B).
- So, we are considering one of Crystal's 9.8 million cable Internet customers (see Table).
- So, the sample space is the Crystal's 9.8 million cable Internet customers.
- Because 6.5 million of these 9.8 million cable Internet customers also have Crystal's cable television service, we have

$$P(A|B) = \frac{6.5}{9.8} = 0.66$$

(The probability of A, given B)

This says that the probability that the randomly selected cable passing has Crystal's cable television service, given that it has Crystal's cable Internet service, is 0.66. That is, 66 percent of Crystal's cable internet customers also have Crystal's cable television service.

Next,

- The randomly selected cable passing has Television service (Given A).
- So, we are considering one of Crystal's 12.4 million cable Television customers (see Table).
- So, the sample space is the Crystal's 12.4 million cable Television customers.
- Because 6.5 million of these 9.8 million cable Internet customers also have Crystal's cable internet service, we have

$$P(B|A) = \frac{6.5}{12.4} = 0.52$$ (The probability of B, given A)

This says that the probability that the randomly selected cable passing has Crystal's internet service, given that it has Crystal's cable Television service, is 0.52. That is, 52 percent of Crystal's cable television customers also have Crystal's cable internet service.

We can see,

$$P(A|B) = \frac{6.5}{9.8} = \frac{6.5/27.4}{9.8/27.4} = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{6.5}{9.8} = \frac{6.5/27.4}{12.4/27.4} = \frac{P(A \cap B)}{P(A)}$$

Here we assume that $P(A)$ and $P(B)$ are greater than 0.

Example 4.10 Gender Issues at a Pharmaceutical Company

At a large pharmaceutical company, 52 percent of the sales representatives are women, and 44 percent of the sales representatives having a management position are women. Given that 25 percent of the sales representatives have a management position, we wish to find:

- The percentage of the sales representatives that have a management position and are women.
- The percentage of the female sales representatives that have a management position.
- The percentage of the sales representatives that have a management position and are men.
- The percentage of the male sales representatives that have a management position.

Define the events:

$Manage \equiv$ the sales representatives that have a management position.

$W \equiv$ the sales representatives that are women.

$M \equiv$ the sales representatives that are men.

$$P(W) = 0.52$$

44 percent of the sales representatives having a management position are women
= 44 percent of the management positions are women.

$$P(W|manage) = 0.44 \qquad P(M|manage) = 0.56$$

$$P(manage) = 0.25$$

- The percentage of the sales representatives that have a management position and are women ($Manage \cap W$).

- The percentage of the female sales representatives that have a management position.

= How much percent of female sales representatives have a management position ($Manage|W$).

= Given: female sales representatives, Find: the probability of management position.

- The percentage of the sales representatives that have a management position and are men ($Manage \cap M$).

- The percentage of the male sales representatives that have a management position ($Manage|M$).

$$P(Manage \cap W) = P(Manage)P(W|Manage) = P(W)P(Manage|W)$$

$$P(Manage \cap W) = P(Manage)P(W|Manage) = 0.25 \times 0.44 = 0.11$$

$$P(Manage|W) = \frac{P(manage \cap W)}{P(W)} = \frac{0.11}{0.52} = \frac{11}{52} = 0.2115$$

$$P(Manage \cap M) = P(Manage)P(M|Manage) = P(M)P(Manage|M)$$

$$P(Manage \cap M) = P(M)P(M|manage) = 0.25 \times 0.56 = 0.14$$

$$P(Manage|M) = \frac{P(Manage \cap M)}{P(M)} = \frac{0.14}{0.48} = 0.2917$$

Note: The probability of the event *MGT* is influenced by whether the event *W* occurs. In such a case, we say that the events *MGT* and *W* are dependent. If $P(Manage|W)$ were equal to *P*(*Manage*), then the probability of the event *MGT* would not be influenced by whether *W* occurs. In this case we would say that the events *MGT* and *W* are independent.

Independent Events

Two events A and B are **independent** if and only if

1. $P(A|B) = P(A)$, or equivalently,

2. $P(B|A) = P(B)$

Here we assume that $P(A)$ and $P(B)$ are greater than 0

Example 4.11 Gender Issues at a Pharmaceutical Company (if time permits)

Recall example 4.10 we see that

$$P(manage) = 0.25$$

25% of the sales representatives are in management

$$P(Manage|W) = 0.2115$$

21.15% of the female sales representatives are in management

$$P(Manage|M) = 0.2917$$

29.17% of the male sales representatives are in management

We can conclude that women are less likely to have a management position at the pharmaceutical company than man.

The Multiplication Rule for Two Independent Events

If A and B are **independent** events, then

$$P(A \cap B) = P(A)P(B)$$

The Multiplication Rule for N Independent Events

If $A_1, A_2, \ldots, A_N$ are **independent** events, then

$$P(A_1 \cap A_2 \cap A_2 \ldots A_N) = P(A_1)P(A_2) \ldots P(A_N)$$

Example 4.12 An Application of the Independence Rule: Customer Service (if time permits)

- This example is based on a real situation encountered by a major producer and marketer of consumer products. The company assessed the service it provides by surveying the attitudes of its customers regarding 10 different aspects of customer service—order filled correctly, billing amount on invoice correct, delivery made on time, and so forth.

- When the survey results were analyzed, the company was dismayed to learn that only 59 percent of the survey participants indicated that they were satisfied with all 10 aspects of the company's service.

- Upon investigation, each of the 10 departments responsible for the aspects of service considered in the study insisted that it satisfied its customers 95 percent of the time. That is, each department claimed that its error rate was only 5 percent.

- Company executives were confused and felt that there was a substantial discrepancy (difference) between the survey results and the claims of the departments providing the services. However, a company statistician pointed out that there was no discrepancy. To understand this, consider randomly selecting a customer from among the survey participants, and define 10 events (corresponding to the 10 aspects of service studied):

$A_1 \equiv$ the customer is satisfied that the order is filled correctly (aspect 1).
$A_2 \equiv$ the customer is satisfied that the billing amount on the invoice is correct (aspect 2).
…
$A_{10} \equiv$ the customer is satisfied that the delivery is made on time (aspect 10).

Also, define the event
$S \equiv$ the customer is satisfied with all 10 aspects of customer service.

Because 10 different departments are responsible for the 10 aspects of service being studied, it is reasonable to assume that all 10 aspects of service are independent of each other. For instance, billing amounts would be independent of delivery times. Therefore, $A_1, A_2, \ldots, A_{10}$ are independent events, and

$$P(S) = P(A_1 \cap A_2 \cap A_2 \ldots A_{10}) = P(A_1)P(A_2) \ldots P(A_{10})$$

If, as the departments claim, each department satisfies its customers 95 percent of the time, then the probability that the customer is satisfied with all 10 aspects is

$$P(S) = (0.95)(0.95) \ldots (0.95) = 0.5987$$

This result is almost identical to the 59 percent satisfaction rate reported by the survey participants.

If the company wants to increase the percentage of its customers who are satisfied with all 10 aspects of service, it must improve the quality of service provided by the 10 departments. For example, to satisfy 95 percent of its customers with all 10 aspects of service, the company must require each department to raise the fraction of the time it satisfies its customers to x, where x is such that $x^{10} = 0.95$. It follows that $x = 0.95^{(\frac{1}{10})}$

So, each department must satisfy its customers 99.49 percent of the time (rather than the current 95 percent of the time).

Thank you!